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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 27: Case Study-II: TOD planning for Belgrade, Serbia

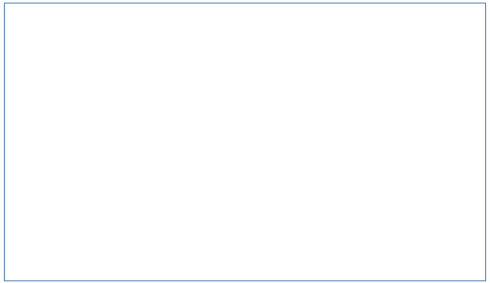
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TOD planning for Belgrade, Serbia

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Study area: Belgrade, Serbia

- Serbia is small sized country situated in Europe.
- Total population of Serbia is approx. 7 million.
- Belgrade is the capital city of Serbia.



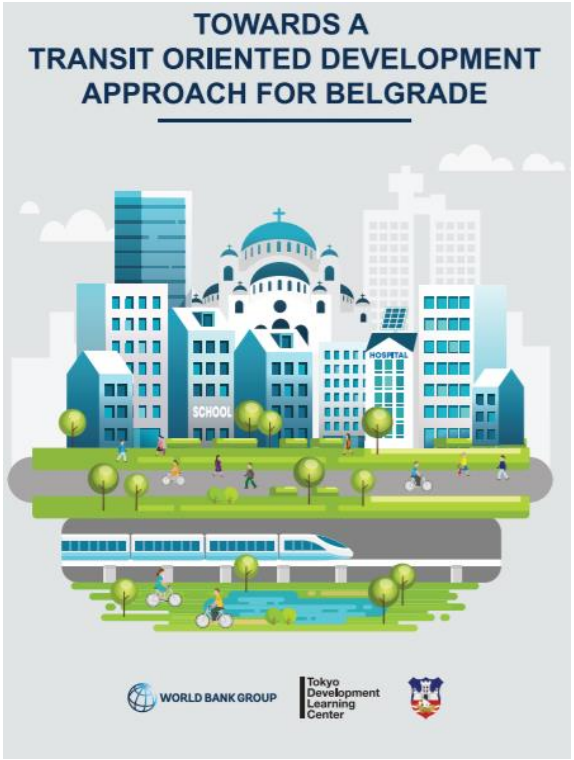


Image:
Location of Serbia on globe (top)

Aerial view of city Belgrade (on left)

Case study report: Planning for TOD in Belgrade, Serbia

- Report is a World Bank document prepared with the help of Tokyo Development Learning Center.
- Report discusses planning process for TOD in Belgrade.



Aim of project

- Assess the opportunities to better integrate land use and public transit planning in Belgrade.
- Develop an approach toward Transit-Oriented Development (TOD) for Belgrade.
- Synergy should be there between transport master plan, which is to be implemented over the next 16 years (2017 to 2033) and land use development planning.

Need of TOD planning

- Very low intensity land use in the areas around transit.
- The stations are treated as just no more than transport nodes without capturing their value for commercial potential.
- No high-density dwellings near stations to provide convenient access to transit and generate much larger transit customer base.
- Thus, lack of synergy between land use planning and public transport planning in Belgrade.

Example: Low intensity development

- Two prominent stations of existing urban rail (shown in image)
- There are no development around these stations (no commercial use of nearby land or residential complex)
- People need to access the station mostly using cars



BG:Voz Novi Beograd Station



BG:Voz Toshin Bunar Station

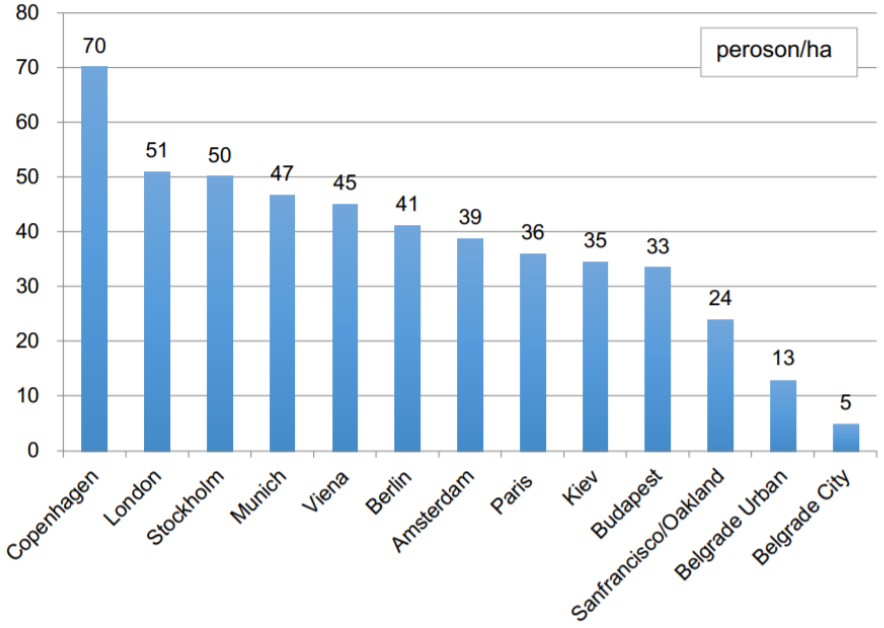
Urban Sprawl in Belgrade

- The total population of Belgrade City was approximately 1.66 million in 2011, growing around 5.3% from 1.58 million in 2002 (Census 2011).
- All three of the fast-growing municipalities are located east of downtown.
- There is a strong trend of population hollowing out from the center to the peripheries within a 10-km radius.



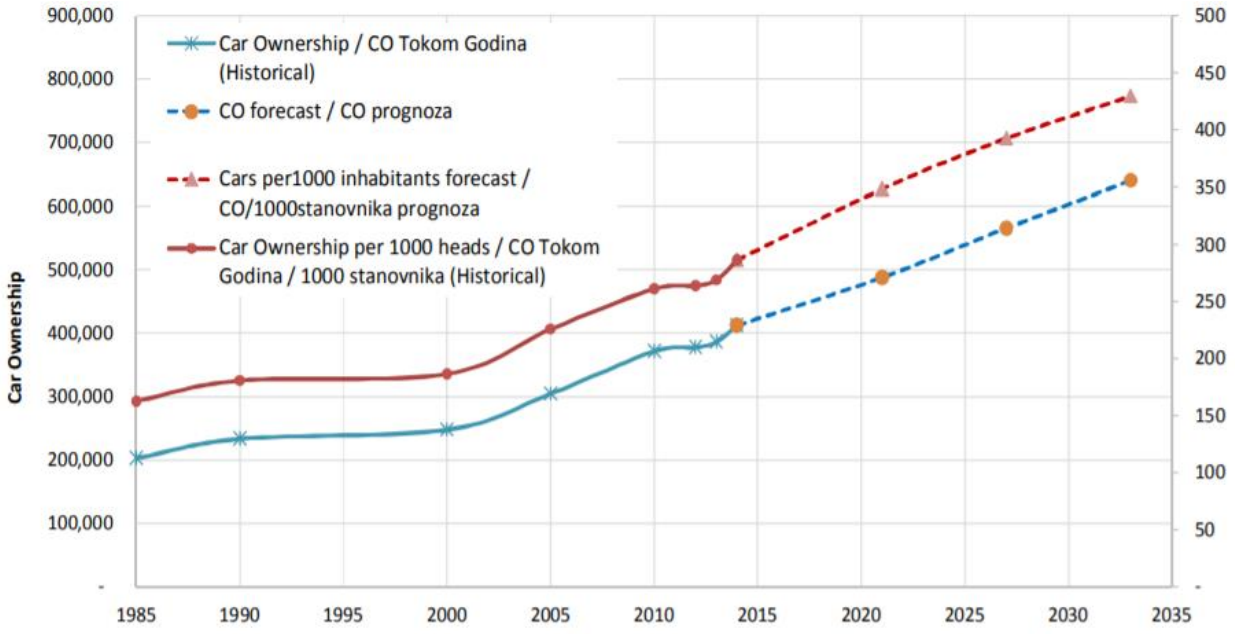
Image source: dreamstime.com

Low density development



- Belgrade has population density of 5 persons/hectare which is very low compared to any major city in Europe
- Low population is leading to dispersed settlement

Car Ownership in Belgrade



- Car ownership data of two largest district in capital city.
- Car ownership forecast suggests possible rapid increase in car ownership by year 2032.

Congestion in Belgrade

- 15% of road junctions will experience extreme congestion in 2033.
- The average volume to capacity ratio is predicted to increase from 0.43 in 2015 to 0.61 in 2033.
- Traditionally 0.5 is considered as a threshold.



Image source: reuters.com

Status of Public Transport (PT) in Belgrade, (2018)

Mode	Daily Passengers (million passenger/day	Daily Passenger-km (million pass-km/day)	Operating Line Distance (km)	Line Passenger Density (passenger/km/day)
Public Buses	1.7	6.2	800	7,750
Trams	0.2	0.6	100	6,000
Trolley Buses	0.16	0.4	-	
BG Voz	27,200	-		
Beo Voz	29,000	-		

Index	Unit	Public Buses	Trams	Trolley Buses	BG:Voz	Beo:Voz
Frequency	min	4-5	5-12	4-16	15	No info

Four different modes of public Transportation

- Public buses
- Trams
- Buses
- suburban rail

Aged population in Belgrade

- One key demographic concern is growth of the aged population.
- In 2002, 22% of the city population was over 60 years old. The same age group grew to 24% in 2011 which will further increase.
- Aged population need support of good public transit to move within the city independently.
- While designing any infrastructure for aged population, it is mandatory to have access to all amenities in nearby locations to transit.

Mode share of Belgrade city

Mode	Mode share (%)
Bus	40
Car	26
Walk	24
Tramway	2
Trolley bus	1.3
Cycle	1
BG train	0.4
Motorcycle	0.3

- Urban buses are the most predominant mode of transport in Belgrade
- Bus share of 40%, followed by passenger cars with a share of 26%

TOD guiding principles for Belgrade 1/2

- Higher Population Density for Public Transport Demand: creating a win-win relationship between transit and land development

- Incentives for Public Transport and Penalties for Automobiles: Promotion and support for public transit and discouraging private automobile using high parking fees, restrictions into central zones etc.

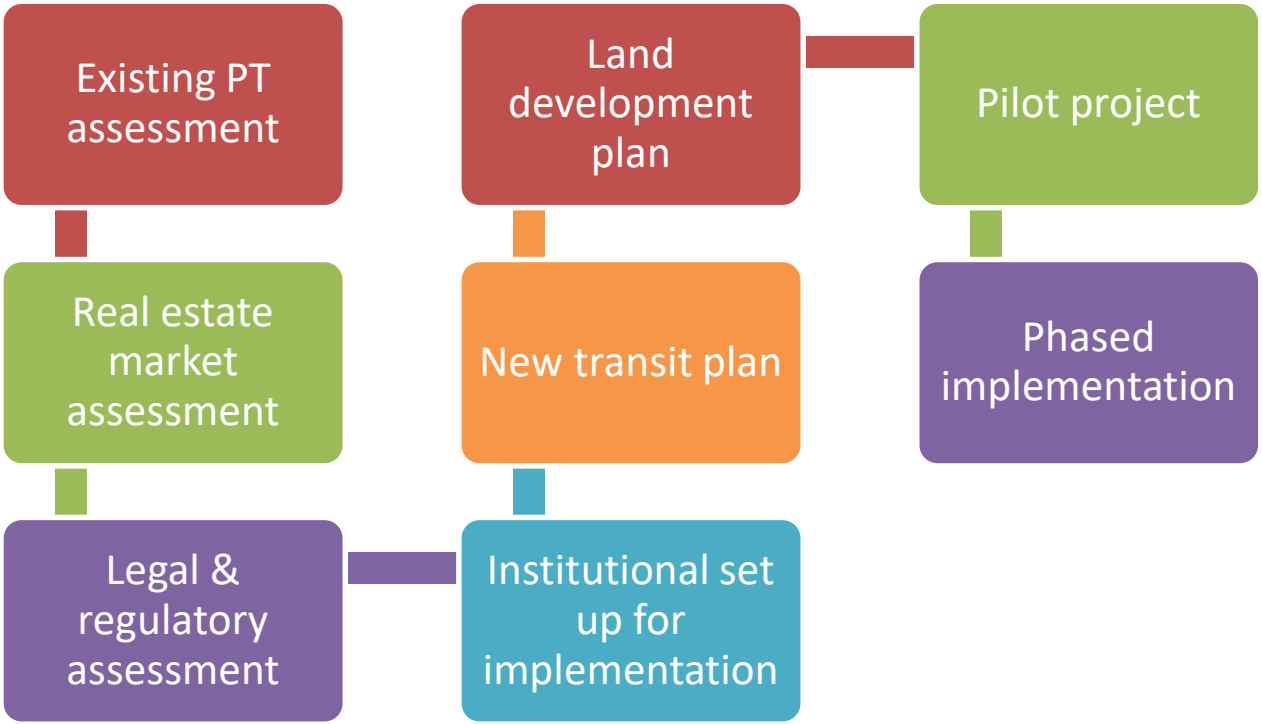
- Non-Automobile Multimodal Alliance: Continuous and user-friendly atmosphere between walking/bicycling to transit stations, proximity and connection routes for NMT

TOD guiding principles for Belgrade 2/2

- Land Value Capture: Creation of system to capture wind-fall (indirect) benefits generated by establishment of transit stations and subsequent population attraction or commercial developments.

- Institutional Barrier Removal: creation of incentives and implementation mechanisms to merge the functions of planning and implementation of transportation and urban development.

TOD process steps



TOD planning process : Public transit assessment

- Assessment of current operation of public transit including organizations, manpower, skill levels, maintenance, and finance
- Customer perspectives by analyzing satisfaction levels as commuting time, waiting, comfort, accessibility, transfer
- Assessment of willingness to pay for services in response to improved services
- Recommendations for service level improvement strategies



Image: Existing urban rail BG-Voz
Source: bgmetro.rs

TOD planning process: Real Estate market assessment

- Assess current trends including demographics and market demand for real estate by segments such as by age group and income class.
- Frameworks including land use control and scope for land value capture.
- Quantification of segmented markets such as high/mid/low class residential, commercial and office.
- Detailed study of areas near existing and planned stations, their ownership, surrounding communities and populations, potential for developments.



Image: One of the posh locality of Belgrade

Image source: culturetrips.com

TOD planning process: Legal & regulatory assessment

- Assessment of legal restrictions and freedom to make financial or institutional plans for any TOD project.
- A cross sector assessment on the national and local legal and regulatory setup should be conducted to grasp the potential and limitations for TOD for Belgrade.



Image: Apex court of Serbia

TOD planning process: Institutional Setup

- Integration of urban transit and development to put all the functions under a single agency.

- This omni-powered agency model to combine profits from land development with transit operation is a straight-forward system to capture the full value of the land.

Integrated Transit Conglomerate Model: a transit company undertaking both land development, commerce and transit services jointly as exemplified by Hong Kong MTR or many Japanese private suburban railway companies with net profitability.

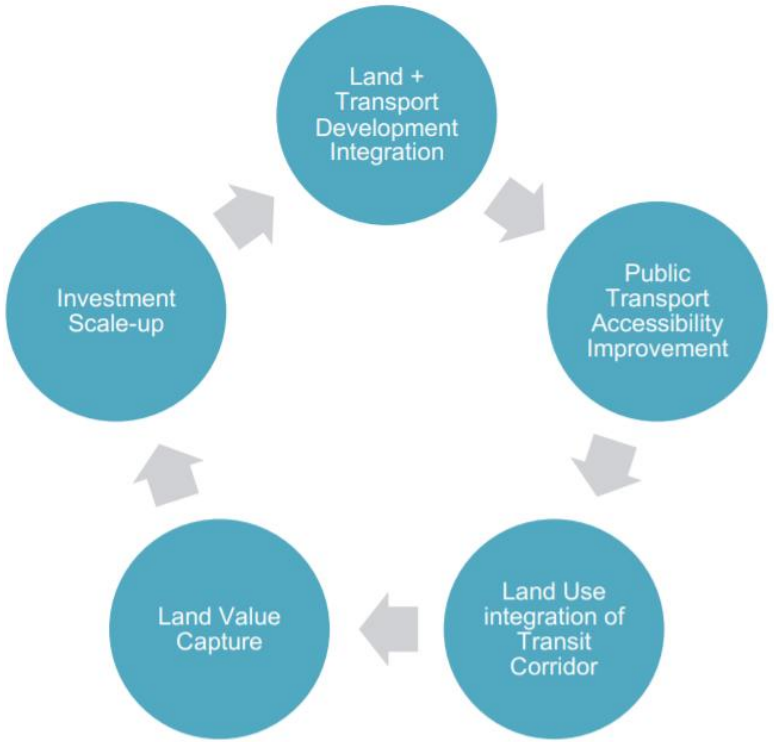
TOD planning process: Pilot Project Plan

- Pilot projects should be designed as integrated plans encompassing both real estate and public transit developments at the same time.
- Pilot projects are small with limited investment.
- Works as model validation for overall project.
- Limitations/drawbacks are recognized and based on this, further project can be scrapped or implemented with improvements.

TOD phased development

- High financial & other resources involved in TOD process and there is always a possibility of project failure.
- To lower the risks, transit and land development projects are implemented in phased manner.
- Generally, initial development is done in areas where success chances are highest.
- This phenomena is known as positive feedback cycle.

Positive Feedback cycle



- First phase investment and development is done.
- Increased accessibility and positive impacts are recognized by residents and decision makers.
- Greater acceptability further boosts the process for next phase process

Plan of transit-1/2

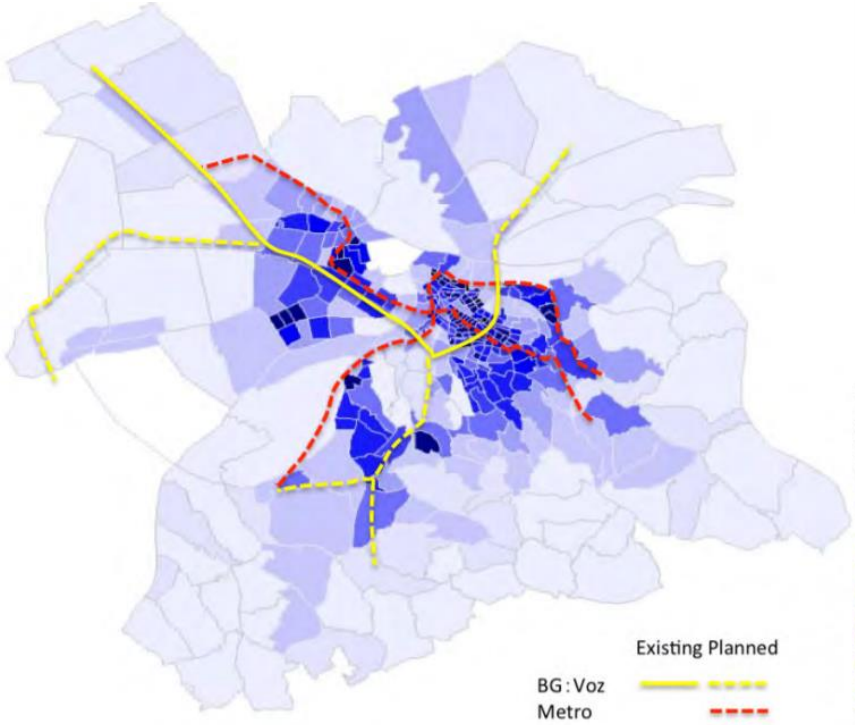


- Two new metro lines are proposed. (Line-L1, Line-L2)
- New transit will cover the areas where public transport is not accessible.

Plan for Transit- 2/2

- Planned transit will include extension of existing urban railway.
- Two metro will cover newly developed areas in capital city and serve the areas which were not connected previously through any PT.

Image: Population density in Belgrade. Darker shade showing higher/dense land use.

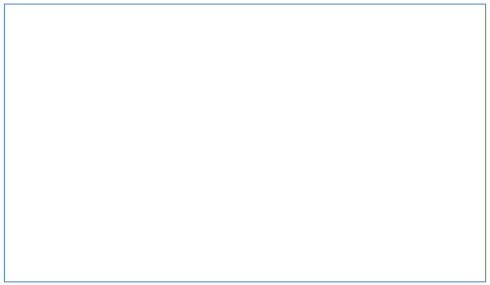




Land development project

- Several large-scale, long-term urban development projects envisaged for Belgrade

- Few prominent projects:
  - Development of the Makis Polje area
  - Waterfront Railway Yard Conversion Project
  - Redevelopment at Ada Huja
  - Waterfront



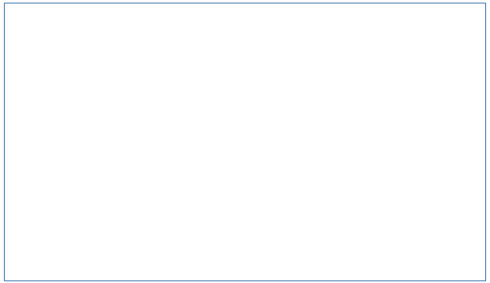
Public Private Partnership (PPP) for TOD

- A tie-up with private investors to develop a commercial building/complex as part of station building
- A joint venture with investors to develop a residential area close to newly planned stations
- Air right sales around stations: the city can sell development rights that allow for higher densities or taller structures beyond the limits specified in land use regulations to raise revenue for public infrastructure and services.

PPP: Best use of expertise and capital of private sector incorporated for public projects through it

Risk sharing in PPP for TOD

- Risks need to be assessed and a mechanism to reduce risks should be designed.
- Land value sharing mechanism from land use planning, density planning and urban plot designs, all commensurate with the expected passenger traffic sizes should be prepared.



Land value capture: Tax/fee-based methods

Property and land tax	Tax levied on estimated value of land or land and buildings combined, with revenues usually going into budgets for general purposes.
Betterment charges and special assessments	Surtaxes imposed by governments on estimated benefits created by public investments, requiring property owners who benefit directly from public investments to pay for their costs.
Tax increment financing	A surtax on properties within an area that will be redeveloped by public investment financed by municipal bonds against the expected increase in property taxes. Mainly used in the United States.
Land sale or lease	Governments sell developers land or its development rights, whose values have increased thanks to a public investment or regulatory charge, in return for and up-front payment, leasehold charge, or annual land rent payments through the term of the lease.

Source: Hiroaki Suzuki, Jin Murakami, et al “Financing Transit-Oriented Development with Land Values”

Land value capture: Development based method 1/2

Joint development	A well-coordinated development of transit station facilities and adjacent private properties between transit agencies and developers, where the latter usually contribute physically or financially to the construction of the station facilities, as their property value will increase thanks to the transit investment. Used in Japan, the United States, and other countries.
Air rights sale	Governments sell development rights extended beyond the limits specified in land use regulations (such as floor area ratios [FARs]) or created by regulatory changes to raise funds to finance public infrastructure and services.

Source: Hiroaki Suzuki, Jin Murakami, et al “Financing Transit-Oriented Development with Land Values”

Land value capture: Development based method 2/2

Land readjustment	Landowners pool their land and contribute a portion of their land for sale to raise funds and partially defray public infrastructure development costs.
Urban redevelopment schemes	Landowners and a developer establish a cooperative entity to consolidate piecemeal land parcels into a single site that they then develop [such as a high-rise mixed use building] with new access roads and public open spaces. The local government modifies zoning codes [typically around rail transit stations] and finances the infrastructure. Mainly used in Japan.

Source: Hiroaki Suzuki, Jin Murakami, et al “Financing Transit-Oriented Development with Land Values”

Provisions for walk & cycle mode

- Currently, walking captures 24% while bicycle captures only 1% of people traveling in urban areas of the city.
- Initially, bicycle lanes focus on linkages to transit stations and parking systems.
- Transit services will capture bicyclers from 2-3 km & will attract pedestrian from 500m radius.
- Biking and walking, well-integrated with public transport planning might increase walk & cycle share sufficiently.

Barrier in implementing TOD: Poor perception for public transport (PT)

- There is a perception in Belgrade that public transport is not the first choice due to its slow speed and poor accessibility especially if one can afford an automobile and parking access in the city center.

- Challenge is to change this perception for social acceptability of project.

Solution:

Bring frequency of existing PT under 5 min.

Provision of good feeder service such as taxis or trams.

Performance based popularity- PT is reliable and punctual mode between the Old City and New Belgrade.

Barrier in implementing TOD: Financial constraints

- The required total investment is estimated to be over 400 million euros excluding the metro project.

- Thus, implementation will require large sums of money for both capital and operational expenses, while the city has limited financing resources.

Solution

- Belgrade city authorities to explore self-financing project schemes for urban transit and land development, incorporating land value capture (LVC).

- Phased manner will decrease initial investment.

Conclusions

- Attributes of Belgrade indicate for coordinated development plan of transit and land use.
- TOD is a possible solution for current and forecasted issues in Belgrade development.
- For making TOD viable, various land value capturing methods should be used.
- An overall institutional set-up should be done for implementing TOD.

References

- Hiroaki Suzuki, Jin Murakami, et al “Financing Transit-Oriented Development with Land Values”
- TCRP report, 2008 “Effects of TOD on housing, parking & travel”
- World Bank, 2018 “Towards a Transit Oriented Development Approach for Belgrade”

Thank You